

Explorer X

Assignments01

Assignment 3

Payment Descriptor: MVC

WS Web Services

Resources

c/main/java

com.controller

EmployeeServlet.java

com.dao

EmployeeDAO.java

com.models

Employee.java

libraries

in

java

webapp

META-INF

WEB-INF

lib

mysql-connector-j-9.6.0.jar

EmployeeForm.jsp

EmployeeList.jsp

Employee.java

EmployeeForm.jsp X

```
1 <%@ page language="java" contentType="text/html; charset=UTF-8"
2   pageEncoding="UTF-8"%>
3
4 <!DOCTYPE html>
5 <html>
6 <head>
7   <meta charset="UTF-8">
8   <title>Insert title here</title>
9 </head>
10 <body>
11   <form action="EmployeeServlet" method="post">
12     Name : <input type="text" name="name"><br>
13     Email : <input type="text" name="email"><br>
14     Department : <input type="text" name="dept"><br>
15     <input type="submit" value="Register">
16   </form>
17 </body>
18 </html>
```

I


```
Employee.java EmployeeForm.jsp *EmployeeList.jsp X
1 <%@ page language="java" contentType="text/html; charset=UTF-8"
2   pageEncoding="UTF-8" import="java.util.*, com.models.Employee" %>
3 <!DOCTYPE html>
4 <html>
5 <head>
6   <meta charset="UTF-8">
7   <title>Employee List</title>
8 </head>
9 <body>
10  <h1>Employee Is Registered In The Company</h1>
11  <a href="EmployeeForm.jsp">Add Another Employee</a>
12  <table border="1">
13    <tr>
14      <th>Id</th>
15      <th>Name</th>
16      <th>Email</th>
17      <th>Department</th>
18    </tr>
19
20  <%
21    List employees = (List) request.getAttribute("Employee");
22
23    if (employees != null) {
24      for (Object obj : employees) {
25        Employee e = (Employee) obj;
26      }
27    <tr>
28      <td><%= e.getId() %></td>
29      <td><%= e.getName() %></td>
30      <td><%= e.getEmail() %></td>
31      <td><%= e.getDept() %></td>
32    </tr>
33  <%
34    }
35  }
```



```
7 <title>Employee List</title>
8 </head>
9 <body>
10 <h1>Employee Is Registered In The Company</h1>
11 <a href="EmployeeForm.jsp">Add Another Employee</a>
12 <table border="1">
13 <tr>
14 <th>Id</th>
15 <th>Name</th>
16 <th>Email</th>
17 <th>Department</th>
18 </tr>
19
20 <%
21 List employees = (List) request.getAttribute("Employee");
22
23 if (employees != null) {
24     for (Object obj : employees) {
25         Employee e = (Employee) obj;
26     }
27 <tr>
28 <td><%= e.getId() %></td>
29 <td><%= e.getName() %></td>
30 <td><%= e.getEmail() %></td>
31 <td><%= e.getDept() %></td>
32 </tr>
33 <%
34     }
35 }
36 %>
37
38 </table>
39 </body>
40 </html>
```




```
Employee.java × EmployeeForm.jsp *EmployeeList.jsp
1 package com.models;
2
3 public class Employee {
4     private int id;
5     private String name;
6     private String email;
7     private String dept;
8
9     public Employee() {
10    }
11
12    public Employee(int id, String name, String email, String dept) {
13        super();
14        this.id = id;
15        this.name = name;
16        this.email = email;
17        this.dept = dept;
18    }
19
20    public int getId() {
21        return id;
22    }
23
24    public void setId(int id) {
25        this.id = id;
26    }
27
28    public String getName() {
29        return name;
30    }
31
32    public void setName(String name) {
33        this.name = name;
34    }
35
36    public String getEmail() {
```


Employee.java X EmployeeForm.jsp *EmployeeList.jsp

```
16     this.email = email;
17     this.dept = dept;
18 }
19
20 public int getId() {
21     return id;
22 }
23
24 public void setId(int id) {
25     this.id = id;
26 }
27
28 public String getName() {
29     return name;
30 }
31
32 public void setName(String name) {
33     this.name = name;
34 }
35
36 public String getEmail() {
37     return email;
38 }
39
40 public void setEmail(String email) {
41     this.email = email;
42 }
43
44 public String getDept() {
45     return dept;
46 }
47
48 public void setDept(String dept) {
49     this.dept = dept;
50 }
51 }
```



```
Employee.java EmployeeForm.jsp *EmployeeList.jsp EmployeeDAO.java X
1 package com.dao;
2
3 import java.sql.*;
4
5
6
7
8
9 public class EmployeeDAO {
10
11     String jdbcURL = "jdbc:mysql://localhost:3306/mca_mvc";
12     String username = "root";
13     String password = "12345";
14
15     Connection getConnection() throws SQLException {
16         try {
17             Class.forName("com.mysql.cj.jdbc.Driver");
18         } catch (ClassNotFoundException e) {
19             e.printStackTrace();
20         }
21         return DriverManager.getConnection(jdbcURL, username, password);
22     }
23
24     public void insertEmployee(Employee e) throws SQLException {
25         Connection con = getConnection();
26         String sqlQuery = "INSERT INTO Employee (name,email,dept) VALUES (?, ?, ?)";
27
28         PreparedStatement ps = con.prepareStatement(sqlQuery);
29         ps.setString(1, e.getName());
30         ps.setString(2, e.getEmail());
31         ps.setString(3, e.getDept());
32
33         ps.executeUpdate();
34         ps.close();
35         con.close();
36     }
37     // New method to get all employees
38     public List<Employee> getAllEmployees() throws SQLException {
39         List<Employee> employees = new ArrayList<>();
40         Connection con = getConnection();
```



```
Employee.java EmployeeForm.jsp *EmployeeList.jsp EmployeeDAO.java EmployeeServlet.java X
1 package com.controller;
2
3 import jakarta.servlet.ServletException;
14
15 @WebServlet("/EmployeeServlet")
16 public class EmployeeServlet extends HttpServlet {
17     private static final long serialVersionUID = 1L;
18
19     EmployeeDAO daoObj = new EmployeeDAO();
20
21     // Handles form submission
22     protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException, IOException {
23
24         String name = request.getParameter("name");
25         String email = request.getParameter("email");
26         String dept = request.getParameter("dept");
27
28         Employee e = new Employee(0, name, email, dept);
29
30         try {
31             daoObj.insertEmployee(e);
32
33             // Get updated list of employees
34             List<Employee> employees = daoObj.getAllEmployees();
35             request.setAttribute("Employee", employees);
36
37             // Forward to JSP to display the list
38             request.getRequestDispatcher("EmployeeList.jsp").forward(request, response);
39
40         } catch (SQLException ex) {
41             ex.printStackTrace();
42             response.getWriter().println("Error: " + ex.getMessage());
43         }
44     }
45
46     // Optional: support direct listing with GET request
```



```
Employee.java EmployeeForm.jsp *EmployeeList.jsp EmployeeDAO.java EmployeeServlet.java X
22 protected void doPost(HttpServletRequest request, HttpServletResponse response) throws ServletException
23
24     String name = request.getParameter("name");
25     String email = request.getParameter("email");
26     String dept = request.getParameter("dept");
27
28     Employee e = new Employee(0, name, email, dept);
29
30     try {
31         daoObj.insertEmployee(e);
32
33         // Get updated list of employees
34         List<Employee> employees = daoObj.getAllEmployees();
35         request.setAttribute("Employee", employees);
36
37         // Forward to JSP to display the list
38         request.getRequestDispatcher("EmployeeList.jsp").forward(request, response);
39
40     } catch (SQLException ex) {
41         ex.printStackTrace();
42         response.getWriter().println("Error: " + ex.getMessage());
43     }
44 }
45
46 // Optional: support direct listing with GET request
47 protected void doGet(HttpServletRequest request, HttpServletResponse response) throws ServletException
48     try {
49         List<Employee> employees = daoObj.getAllEmployees();
50         request.setAttribute("Employee", employees);
51         request.getRequestDispatcher("EmployeeList.jsp").forward(request, response);
52     } catch (SQLException e) {
53         e.printStackTrace();
54         response.getWriter().println("Error: " + e.getMessage());
55     }
56 }
57 }
```


← → ↻ ⓘ localhost:8080/MVC/EmployeeForm.jsp

Name :

Email :

Department :

Employee Is Registered In The Company

[Add Another Employee](#)

Id	Name	Email	Department
1	Arfaat Patel	patelarfaat0@gmail.com	MCA
2	zaid shaikh	zaid@gmail.com	MCA


```
Employee.java EmployeeForm.jsp *EmployeeList.jsp EmployeeDAO.java X EmployeeServlet.java
26 String sqlQuery = "INSERT INTO Employee (name,email,dept) VALUES (?, ?, ?)";
27
28 PreparedStatement ps = con.prepareStatement(sqlQuery);
29 ps.setString(1, e.getName());
30 ps.setString(2, e.getEmail());
31 ps.setString(3, e.getDept());
32
33 ps.executeUpdate();
34 ps.close();
35 con.close();
36 }
37 // New method to get all employees
38 public List<Employee> getAllEmployees() throws SQLException {
39     List<Employee> employees = new ArrayList<>();
40     Connection con = getConnection();
41     String sqlQuery = "SELECT * FROM Employee";
42     PreparedStatement ps = con.prepareStatement(sqlQuery);
43     ResultSet rs = ps.executeQuery();
44
45     while (rs.next()) {
46         Employee e = new Employee(
47             rs.getInt("id"),
48             rs.getString("name"),
49             rs.getString("email"),
50             rs.getString("dept")
51         );
52         employees.add(e);
53     }
54
55     rs.close();
56     ps.close();
57     con.close();
58
59     return employees;
60 }
61 }
```